



WATERLILY LABS

Great Logics, Great Solutions

C# WEB FULL-STACK DEVELOPMENT TEST -2

We're very excited to have you! Read the following brief about this assignment and show us how good you are. Good luck!

INTRODUCTION

To demonstrate your C# web development skills, you are required to create a ASP.NET Core MVC project.

Please read carefully and understand the requirements before continuing.

ASSESSMENT

You are required to build a simple ASP.NET Core MVC web application with two pages (Product List, Report) for managing a list of products. Each product has a Product ID, Name, Category, Price, and Stock. Additionally, implement calculations to determine:

1. The average price of products in each category (No need to consider category table).
2. The highest stock value category.

Requirements

1. .NET Core Project:

- The project should be an ASP.NET Core MVC application.

2. CRUD Operations:

- Implement create, read, update, and delete operations for a product entity.

3. Service and Repository Layer Pattern:

- Repository Layer: Handle data access and interact with the database.
- Service Layer: Handle business logic and act as an intermediary between the controller and repository layers.

4. Code-First database pattern:

- Use migrations.
- Use stored procedures for your calculations.
- Use MSSQL

5. Caching:

- Implement caching to improve performance, especially for data retrieval operations.
- CachedLong : Cache a generic result until cache is removed (don't worry about removing cache function).

- Cached: Cache a generic result up to 5 minutes.
- Enable the caching mechanism in the service layer.

6. Delegates:

- Use delegates for implementing callback methods.

7. UI / Scripting:

- Use of bootstrap/custom own css.
- Use JQuery.

8. Unit Testing:

- Use MSTest framework
- Write unit tests to verify the correctness of the service layer CRUD operations.
- Use mock objects to isolate tests from external dependencies (e.g. database).

9. Please comment/explain your code well.

BEFORE YOU START:

Please make a project plan document with:

1. Write your understanding of the requirement.
2. Write how you are planning to solve the problem.
3. Identify and note any issues that may arise during the development.
4. Make a pseudo code to cover the required logic.
5. Develop the project according to your own project plan.

WHAT WE EXPECTING:

Complete as much as possible and your work will be assessed by:

1. Coding efficiency.
2. C# knowledge (feel free to show off your coding skills, OCDs).
3. Logics in general how you approach a problem.

QUICK FACTS:

1. Don't use external code, if you do, make sure you understand it well.
2. If you wish, you can split the repository and services into a separate project.
3. Avoid db operations in controller / views.
4. Avoid using templated code from online or lazy codes.

NOTES

- Please do not take much time to complete this assessment period of 3-4 days. If you need more time, please let us know.
- This assessment will remain confidential to Waterlily Labs and will only be viewed internally for hiring and interview purposes only. This submission will not be shared externally.
- If you fail to submit or communicate after the deadline, we assume you are no longer interested in the position or your skills do not match your CV.

CLARIFICATIONS & HELP:

1. Zip the project and share using Dropbox/OneDrive/ Google Drive if it's more than 10M.
2. If you need further clarification or help, please do not hesitate to contact us.

Good luck!

If you need any assistance or have clarifications with the assigned tasks, please feel free to contact us at your convenience.